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Abstract

This Master's Thesis presents a comprehensive study on quantum error correction (QEC), specifically focused on 2D topological color codes. Leveraging state-of-the-art simulation tools such as `stim` and `sinter`, alongside specialized libraries like `ldpc` and `pymatching`, we analyze decoder performance on 4.8.8 lattice geometries. The study concludes with a comparative analysis of the decoding results and explores potential improvements for the error correction process.

This work provides a systematic and structured framework to evaluate the performance of QEC codes and decoders. By implementing a selected subset of these algorithms within an extensible architecture, allowing for future expansions. Furthermore, this manuscript is designed to serve as an accessible introduction for researchers and students entering the field.

QEC DECODERS FOR 2D COLOR CODES

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Master´s Degree in Quantum Computing

Higher Polytechnic School

Quantum Error Correction Decoders for 2D Color Codes: Systematic Simulation and Comparison

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